# Muscle Coactivation Levels Reflect a Tradeoff Between Metabolic Cost and Performance When Responding to Physical Disturbances During Upper Limb Posture Control

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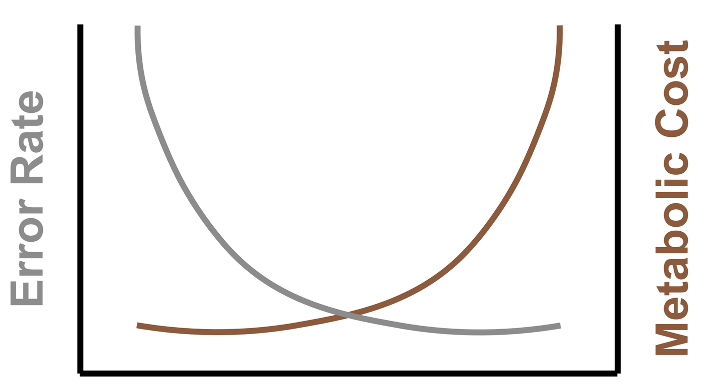
## Introduction

Muscle coactivation - the simultaneous activity of agonist and antagonist muscles - has been shown to improve motor performance by enabling muscles to generate forces more rapidly and respond more effectively to sensory feedback. However, this enhanced performance comes at a cost, and the metabolic expense associated with muscle coactivation remains unknown in the context of upper limb posture control. The main objective of this study is to investigate the trade-off between metabolic cost and performance under different levels of muscle coactivation.

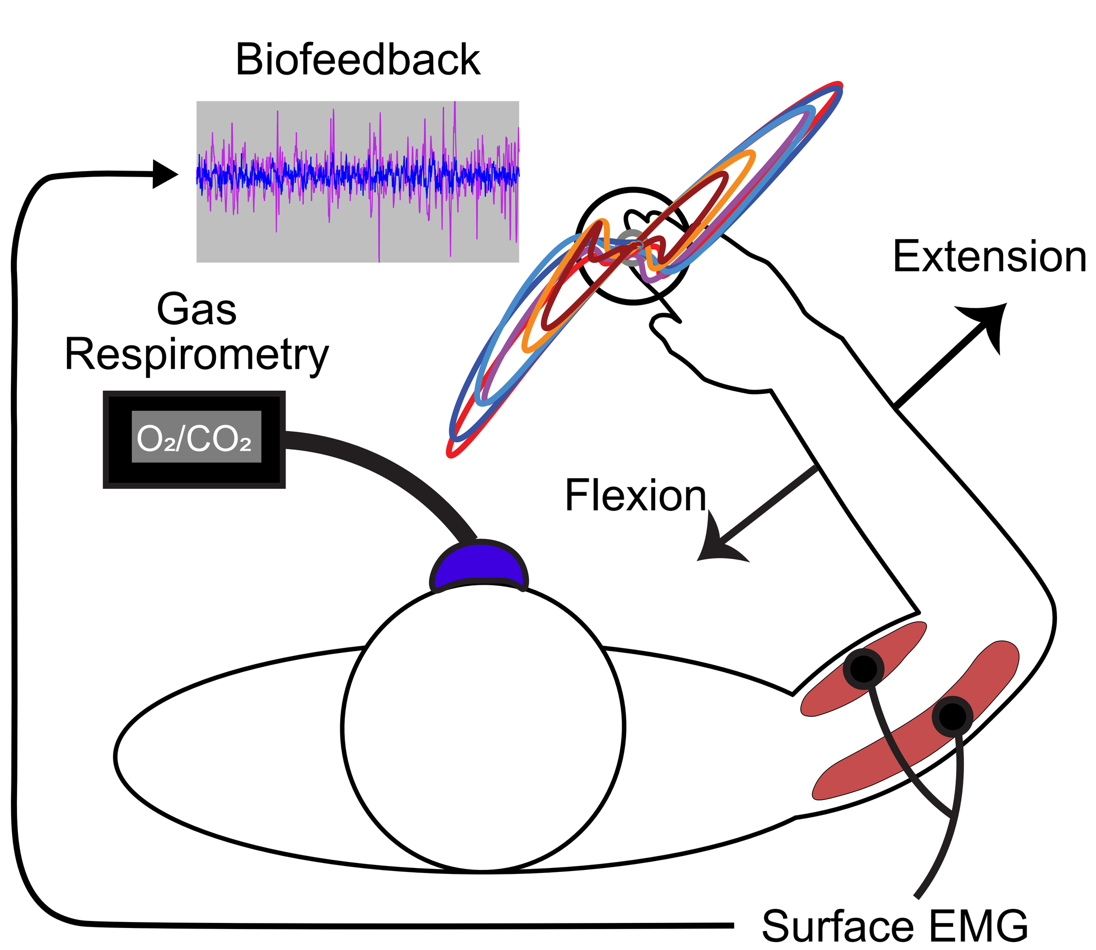
## Hypothesis & Prediction

We hypothesize that there is a trade-off between muscle coactivation, metabolic cost, and task performance. We predict that higher levels of coactivation will lead to improved performance at the expense of increased metabolic cost.




## Methods

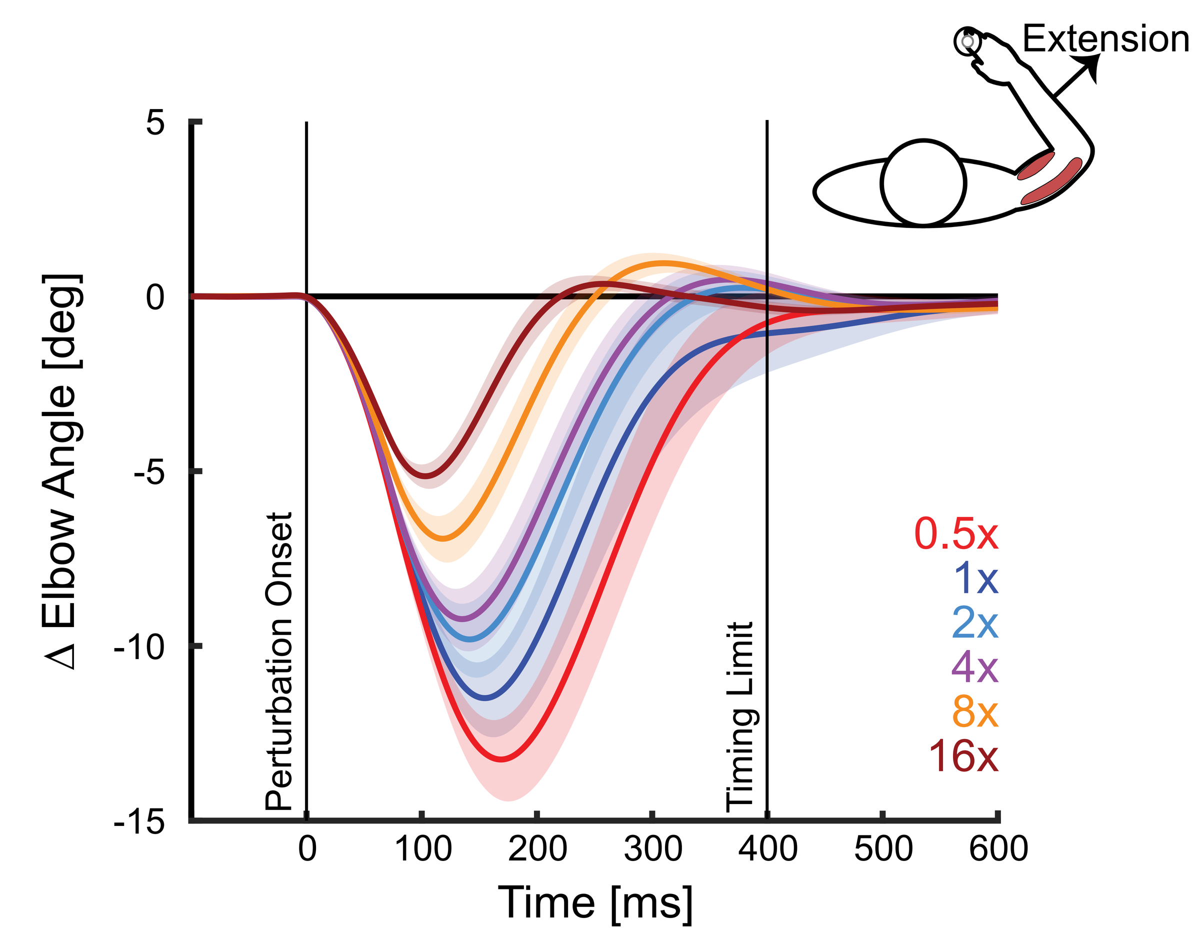
Twelve healthy participants attempted to maintain a fixed arm posture in a Kinarm exoskeleton robot while encountering mechanical disturbances.




Participants performed the task under self-selected coactivation levels. Biofeedback was used to maintain coactivation at [0.5, 2, 4, 8, 16]x their self-selected level, with testing continued until failure.

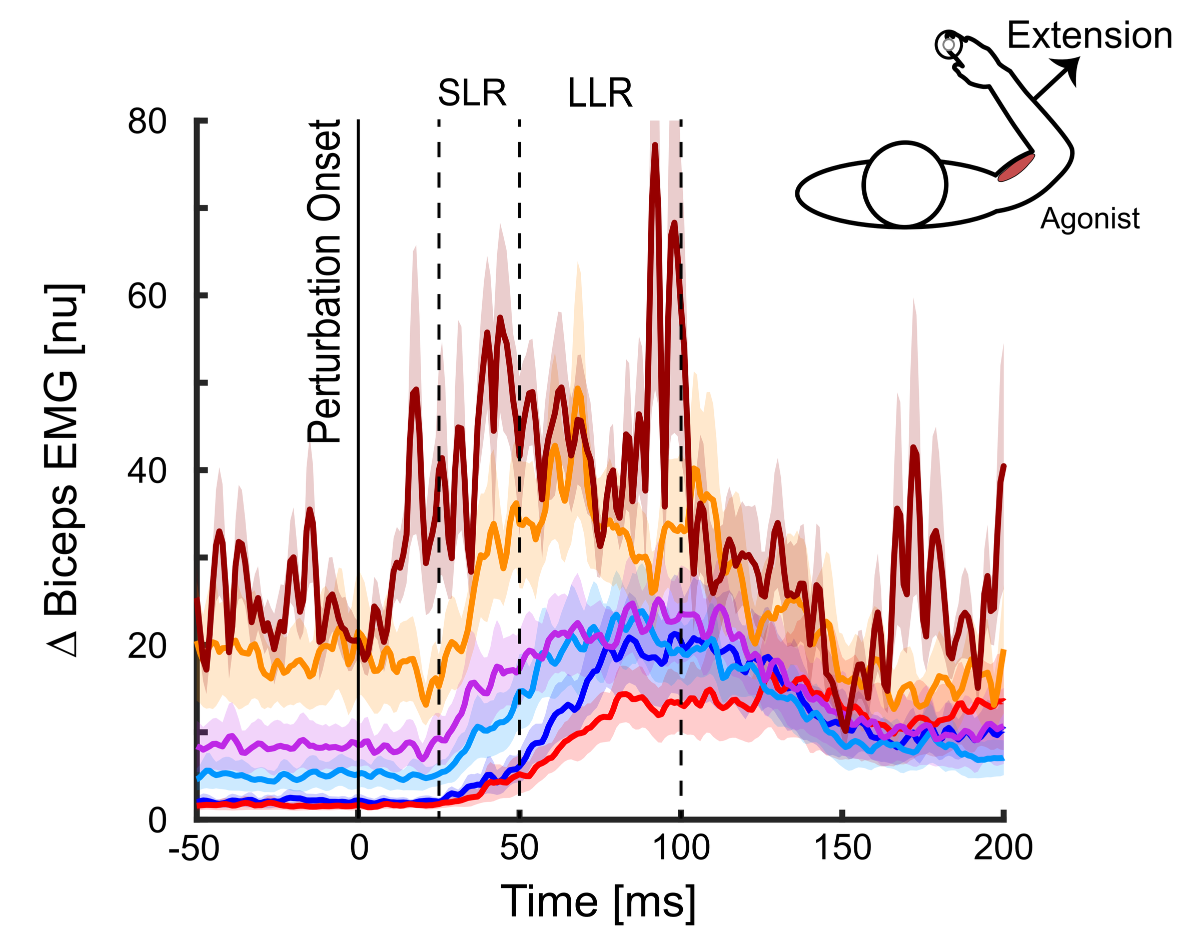
## Results

### Coactivation Reduces Peak Displacement

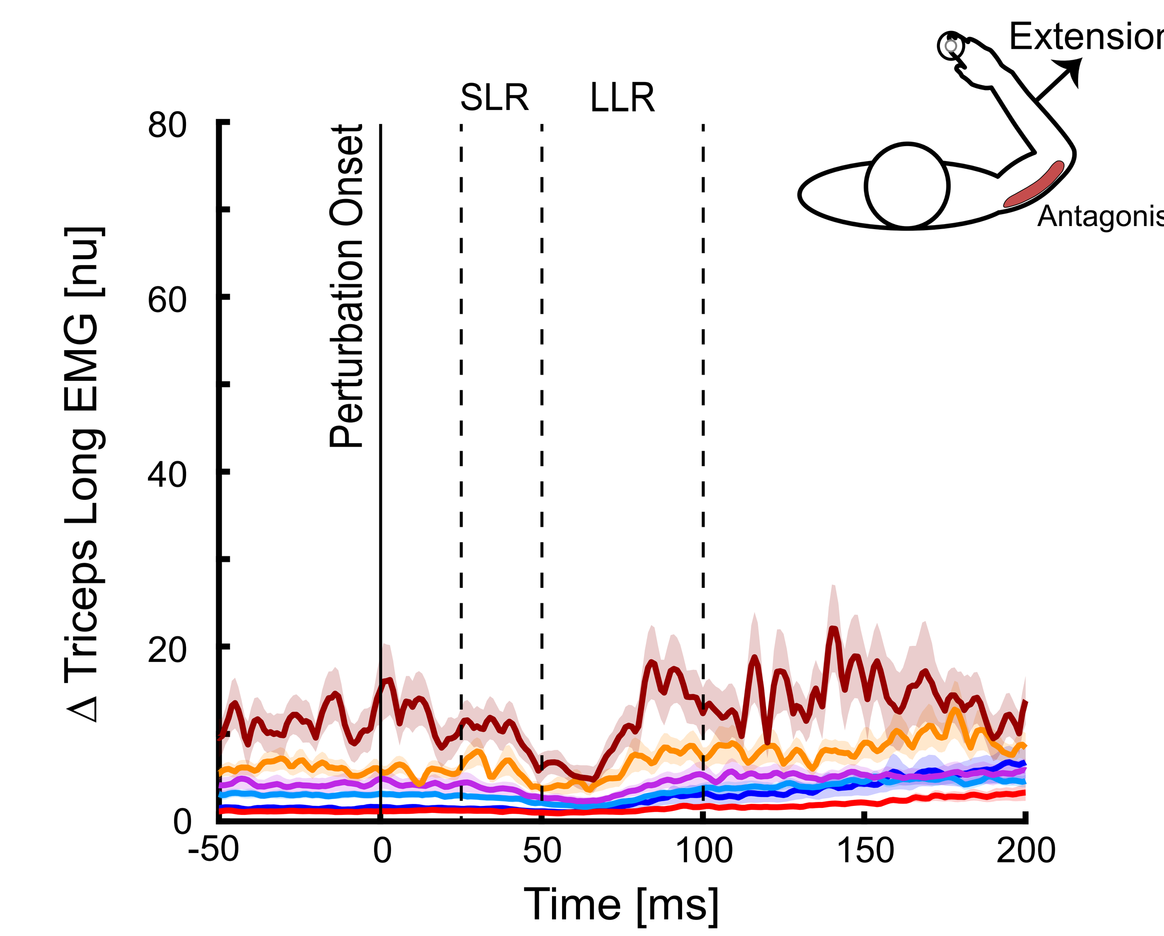



Peak displacements scale with the level of muscle coactivation.

### Muscle Coactivation Increases Responsiveness to Proprioceptive Feedback

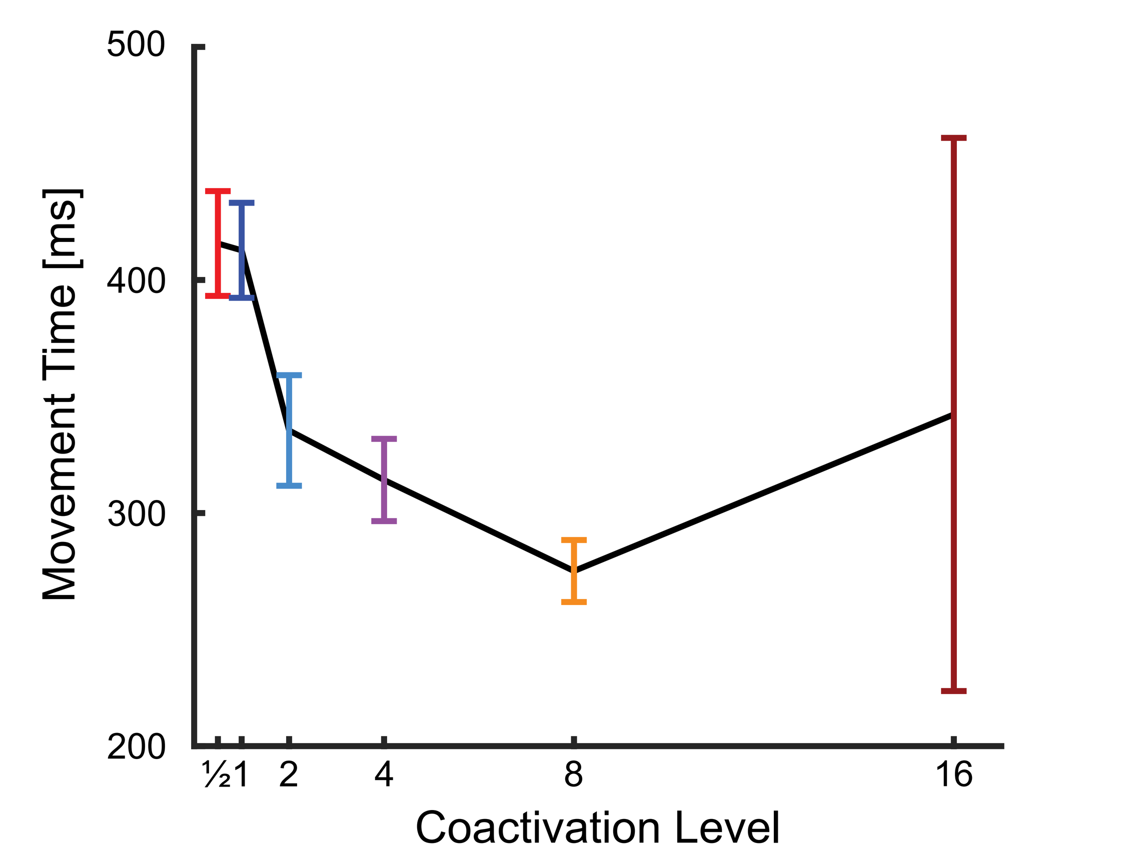



Increased muscle coactivation results in greater stretch responses of agonist muscle.



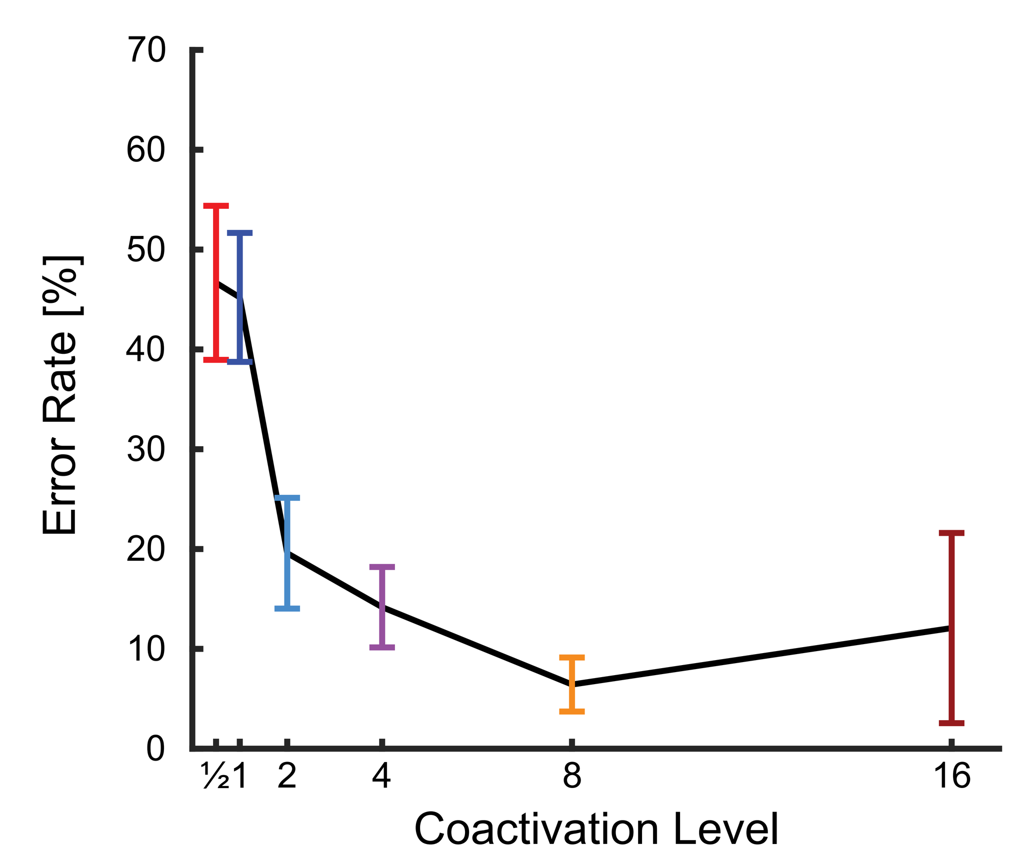

Increased muscle coactivation enables more inhibition of antagonist muscle.

### Coactivation Facilitates Faster Movements

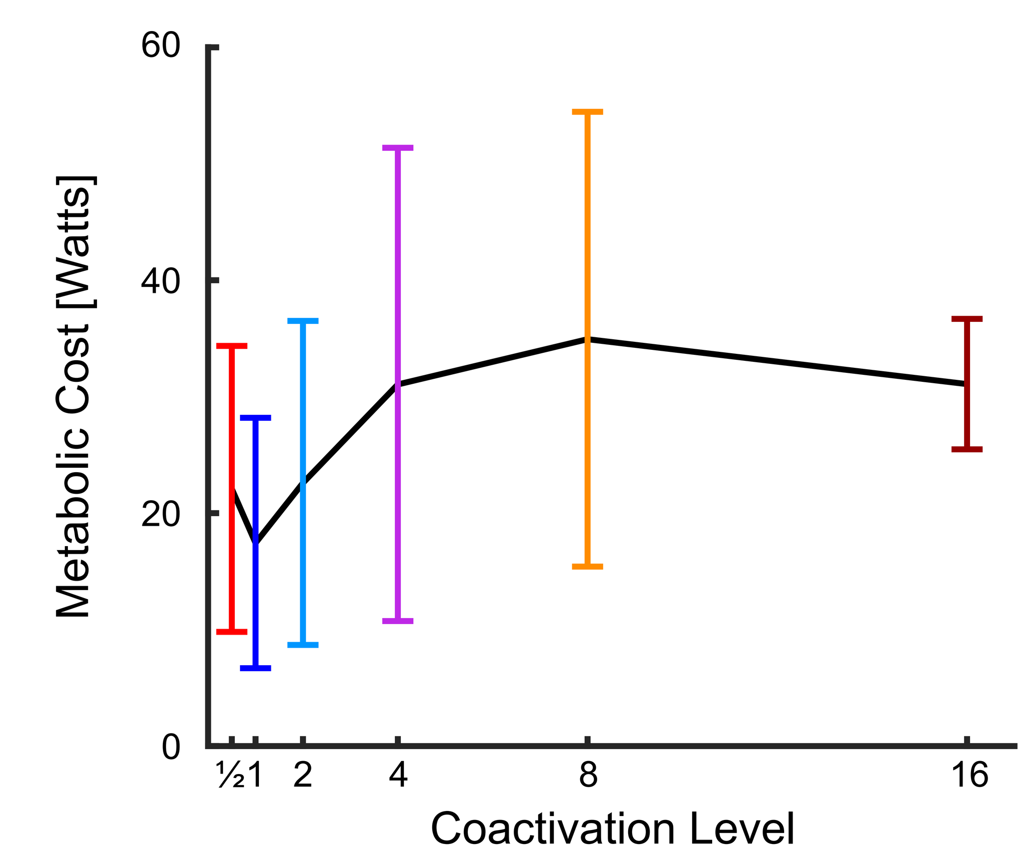



Higher muscle coactivation reduces movement time after perturbation.

### High Coactivation Improves Performance But Increases Metabolic Cost




Higher muscle coactivation reduces error rates in disturbance responses.




Higher muscle coactivation levels lead to increased metabolic cost.

## Discussion

At spontaneous levels of muscle coactivation, the nervous system prioritizes the minimization of metabolic cost at the expense of performance. However, increasing the level of coactivation via biofeedback-controlled muscle activity improves performance at the expense of metabolic cost. This research offers insights into the role of muscle coactivation in responding to sensory feedback.

## Conclusion

There is a complex interplay between metabolic cost and performance in the control of human arm movements and postures.